Dr.Najva Nambrath

Generative AI Projects Developer | LLM & Machine Learning Specialist | Assistant Professor | AI/ML Trainer| AI &Robotics Researcher | Speaker | Author| Reviewer

About Me

Accomplished Generative AI and Machine Learning Scientist with 6 years of specialized experience in developing and deploying advanced AI-driven solutions across academia and industry. Specialized in Large Language Models (LLMs), GenAI frameworks (LangGraph, LangChain), and robotics powered by artificial intelligence. Demonstrated expertise in end-to-end data preprocessing, algorithm development, and the integration of AI into complex robotic and automation systems. Recognized for excellence in research, teaching, and thought leadership, with a commitment to advancing the frontiers of AI, robotics, and data science.

Core Competencies

- Generative AI (GenAI) Systems & LLMs
- LangGraph & LangChain Project Development
- Data Preprocessing, Analysis & Visualization
- Robotics & Autonomous Systems using AI
- Machine Learning & Deep Learning Algorithms
- Computer Vision & Predictive Analytics
- Prompt Engineering
- Academic Instruction & Curriculum Design
- Research Publication & Peer Review

Education

- Ph.D. – Robotic Control Systems using AI
- M.Tech– Communication & Signal Processing
- B.Tech – Electronics &Communication

Technical Skills

- Programming: Python, MATLAB, Simulink, C++
- GenAI & LLM Frameworks: LangGraph, LangChain, Hugging Face, OpenAI APIs
- ML/DL Frameworks: TensorFlow, PyTorch, Keras, Scikit-Learn
- Data Science Tools: Pandas, NumPy, OpenCV, Matplotlib, Seaborn
- Robotics: RoboDK, AI-driven Robotics
- Other: Prompt Engineering, Syllabus Development

Certifications

- Python for data science, AI and Development
- Databases and SQL for Data Science
- Data Science Orientation
- Generative AI Fundamentals
- Data Visualization with Python
- Data Analysis using Python
- Computer vision for faces

Faculty Development Programs

- Spreadsheet skill for researchers
- Nonlinear Systems: Dynamics and Control
- System Modelling and Control Methods

Professional Experience

Generative AI Solutions Architect & LLM Specialist (Freelance AI/ML Developer, 2024– Present)

- Led the design and deployment of advanced GenAI solutions utilizing LLMs, LangGraph, and LangChain for education, automation, and business intelligence.
- Developed robust data preprocessing pipelines, ensuring high-quality, reliable input for AI models and downstream analytics.
- Engineered multi-agent GenAI systems for adaptive instruction, workflow automation, and domain-specific conversational agents.
- Integrated LLMs and GenAI frameworks into real-world applications, optimizing performance and scalability.
- Developed an adaptive AI teaching assistant using LLMs, LangGraph, and LangChain for personalized education.
- Conversational Travel Agent: Built a domain-specific travel assistant leveraging LLMs and prompt engineering for itinerary planning and customer support.

Robotics & AI Researcher (AICTE, 2018-2022)

- Pioneered the application of AI and machine learning in robotic navigation, control systems, and industrial automation.
- Developed and implemented AI-powered robotic solutions using RoboDK, Python, and MATLAB, enhancing operational efficiency and autonomy.
- Supervised research projects on robotics and automation, guiding teams in the development of innovative, AI-driven robotic systems.
- Engineered AI-powered navigation and control algorithms for autonomous robotic platforms.

Machine Learning Scientist (AICTE, 2018-2022)

- Designed and deployed machine learning and deep learning models for predictive analytics, computer vision, and management systems.
- Led multidisciplinary teams in identifying operational challenges and implementing AI-based solutions for industrial automation.
- Conducted comprehensive data preprocessing, analysis, and visualization to extract actionable insights from large, complex datasets.
- Designed scalable, automated pipelines for data cleaning, transformation, and feature engineering in ML projects.

Assistant Professor & AI/ML Trainer ((Government Engineering College, Thrissur, 2022-2023)

- Delivered undergraduate and graduate courses on Machine Learning, Deep Learning, Computer Vision, and Generative AI.
- Developed and authored a comprehensive syllabus for "Introduction to Generative AI".
- Mentored students and professionals, fostering hands-on learning and innovation in AI and robotics.
- Reviewer & Author in Conferences & Journals
- Published research in SCI, Scopus-indexed journals, and IEEE conferences on AI-driven robotics, and deep learning.
- Served as a peer reviewer, upholding rigorous standards for research quality in AI, ML, and robotics.

Journals

- Real time obstacle motion prediction using neural network based extended Kalman filter in robotic path planning, Kuwait journal of Science, 2022
- Neural network based adaptive controller for WMR trajectory tracking, IEEE ACCESS, 2022
- Neural network based TID controller for WMR trajectory tracking, Lecture notes in networks and systems ,2022
- Model reference controller approach for robot arm tracking using neural networks, Indian Journal of science and technology, 2019
- SIFT and tensor based object detection images using deep learning, Procedia Computer Science, 2016

International\National Conferences

- Analysis of trajectory tracking controllers for wheeled mobile robots, IEEE Industrial Electronics and Applications Conference, 2021
- Robot arm movement and trajectory control using neural networks, International conference on space and defense Technologies, 2019
- Neural network based TID controller for WMRs trajectory tracking, International Congress on Information and Communication Technology, 2021
- Review of path planning algorithms, Conference of the Kerala state technological congress , 2019
- SIFT and tensor based object detection videos using DL, International Conference on Advances in Computing & Communications
- SIFT and tensor based object classification in images, International Conference on Information Science, 2016
- LPC based low dimensional features for object classification, International Conference on Communication System, Computing and IT Applications , 2016

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